

CLO #1: Investigate a transformer circuit using Faraday's Law and electric circuit theory

Q1:

[marks 20]

A 60000VA, 4000/230V single phase distribution transformer has series impedance referred to the primary of $12 + j45 \Omega$. The components of the excitation branch referred to the secondary side are $R_C = 50 \Omega$ and $X_M = j 60 \Omega$. Assume a secondary current of 200A with a lagging PF of 0.7547 is provided to a load at rated secondary voltage.

- Draw circuit model referred to secondary side
- Investigate the circuit to find out Voltage regulation
- Find out applied primary voltage
- Find out real power input to the transformer

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Q2:

[marks 5+5+5+5=20]

- Draw B-H curve for DC excitation for a ferromagnetic material, clearly indicate knee point, saturation portion and linear portion
- Name two types of losses that occur in a transformer's core.
- Draw phasor diagram of a transformer for capacitive load
- Draw the schematic diagram to perform short circuit test on a transformer.

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CLO # 2: Investigate working of a DC Machine

Q1:

[marks=25]

A dc series machine is connected across a 240-volt line. It rotates at 1200 rpm and is generating 230 volts. The armature current is 40 amps. **Investigate** the dc machine to find out the following quantities assumed rated terminal voltage in all cases.

- (a) Is the machine functioning as a generator, or as a motor?
- (b) Resistance of the armature circuit.
- (c) Power loss in the armature circuit resistance and the electromagnetic torque in newton-meters.
- (d) Terminal characteristics of above DC series machine.
- (e) General torque-speed curve of above DC series machine.

CLO # 3: Analyze Synchronous Generator performance along with special emphasis towards environmental cost of generation

Q2:

[marks= 25]

A 3 ϕ , 120 MVA, 12 kV, 60 Hz, two-pole, 0.85 lagging PF, Y-connected, steam turbine-driven alternator has a stator resistance of $R_a = 0.015 \Omega$ and a synchronous reactance of $X_s = 0.85 \Omega$. Ignore rotational and mechanical losses. **Analyze** the synchronous generator circuit to find out the following.

- (a) Synchronous speed. Also state the type of rotor.
- (b) Internally generated voltage if the alternator delivers power to an infinite bus at the rated condition. Also draw the phasor diagram.
- (c) At full-load (rated) condition, the efficiency is 92%. At this condition, determine
 - (i) The power lost in the armature resistance.
 - (ii) The rotational loss.
 - (iii) The torque in newton-meter applied to the shaft by the steam turbine prime mover.

$$P_{\text{conv}} = P_{\text{out}} + P_{\text{loss}}$$

$$143.79 - 8142.2$$